

# András Gémes

[shadowshell.io](https://shadowshell.io) | [github.com/gemesa](https://github.com/gemesa) | [linkedin.com/gemesa](https://linkedin.com/gemesa) | [gemesa@protonmail.com](mailto:gemesa@protonmail.com)

## Summary

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Security researcher with 8 years of experience. [Hands-on experience](#) in mobile application security, reverse engineering, malware analysis and vulnerability research. Certified in [Sec+](#), [CASP+/SecX](#), [CEH](#) and [others](#). LLVM compiler engineering (code obfuscation) and embedded systems background. Open to roles in security research.

## Work experience

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**Mobile Security Analyst @ SGS Brightsight - Delft, Netherlands** **June 2026 – Present**

- Analyzing mobile payment applications for vulnerabilities (cryptography, runtime protection, data protection, etc.)
- Evaluating mobile payment applications against the PCI MPoC standard
- Reverse engineering app binaries (Ghidra, Frida) to identify logic flaws and develop PoC exploits

**Security Researcher @ HighTec EDV-Systeme GmbH - Budapest, Hungary** **Feb 2023 – June 2026**

- Evaluated the security impact of reported LLVM vulnerabilities as a [member of the LLVM security group](#)
- Analyzed Windows, Linux and Android [malware](#), writing [detection rules](#) and config extractors ([Ghidra](#), [Python](#))
- Implemented custom [LLVM-based C++ code obfuscation pass plugins](#) based on malware research

**Application Security Engineer @ Knorr-Bremse R&D Center - Budapest, Hungary** **May 2018 – Jan 2023**

- Implemented static (Clang-Tidy) and dynamic (Valgrind) application security testing across embedded C codebases
- Used AFL++ fuzzing and code review to discover embedded systems memory corruption vulnerabilities and build PoC exploits
- Built iOS and Android apps for real-time vehicle data visualization

## Technical skills

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**Reverse engineering (static):** DiE, Ghidra, IDA, ILSpy, dnSpy, Binwalk, Apktool, jadx, otool, ipsw

**Reverse engineering (dynamic):** x64dbg, Sysinternals, LLDB, GDB, Frida, ADB, QEMU, Qiling, VirtualBox, strace

**Detection engineering:** YARA, Sigma, Suricata, capa

**Programming languages:** C, C++, Rust, Objective-C, Swift, Python 3, Java, Assembly (ARM64, x86-64), Bash

**Vulnerability research:** AFL++, checksec, ROPgadget, Clang-Tidy, Valgrind

**Network and traffic analysis:** Wireshark, Burp Suite, mitmproxy, Zeek, FakeNet-NG, INetSim

**Platforms and DevOps tools:** Linux (Fedora, Ubuntu), macOS, Windows, Git, Docker, GitHub Actions, Jenkins

## Certifications

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[CompTIA Security+](#), [CompTIA CASP+/SecurityX](#), [EC-Council CEH](#), [TCM Security PMAT](#), [Invoke RE IMBT](#) and [others](#).

## Open source contributions

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- [ghidra](#): contributing bug reports and patches to Ghidra, focusing on the BSim, Debugger and FunctionID features
- [threat-detection-rules](#): publishing YARA, Sigma and Suricata rules for analyzed malware families
- [ghidra-scripts](#) | [reversing-scripts](#): developing custom scripts to support reverse engineering
- [phantom-pass](#) | [o-mvll](#): implementing and improving LLVM-based code obfuscation pass plugins
- [rust-arm64](#): writing a Rust book about analyzing Rust-to-ARM64 compilation

## Education

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**MSc in Mechatronics Engineering @ [Budapest University of Technology and Economics](#)** **Feb 2016 – June 2018**

**BSc in Mechatronics Engineering @ [University of Pannonia](#)** **Sept 2012 – Jan 2016**